```
function [Ngroup, Np_group, arc_group, N_arc_avg] =
ps_spatial_unwrapping_grouping(A)

% The function is used to group arcs, i.e. subdivide all arcs to a number
% of groups. The number in each group should not larger than the number of
% number of pscs.
%
% Input:    - A          design matrix which is sparse
%
% Output:   - Ngroup     number of groups in total
%           - Np_group   number of points per group
%           - arc_group  number of arcs per group
%           - N_arc_avg  average number of arcs per group
%
% ----------------------------------------------------------------------
% File.............: ps_spatial_unwrapping_grouping.m
% Version & Date..: 1.7.2.16, 12-DEC-2009
% Author...........: Shizhuo Liu
%                   Delft Institute of Earth Observation and Space Systems
%                   Delft University of Technology
% ----------------------------------------------------------------------
%
% This software is developed by Delft University of Technology and is
% intended for scientific use only. Applications for commercial use are
% prohibited.
%
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%
% Change log
%

global Nworkers unwrapping_mode
unwrapping_mode = 'sequential';
Nworkers=[];

if (numel(A) <= 3*10^4) ;   % A*Rinv is small, memory can hold
                            % the number is user dependent, based on
                            % experience, set it to smaller number might
                            % result in slowing down the process, set it
                            % to a too large number might run out of memory

  Ngroup = 1 ;
  Np_group = size(A,2) ; % all pscs
  arc_group{1} = 1 : size(A,1) ; % all arcs
  N_arc_avg = [] ;

  fprintf(1,'The network has %4.0f arcs and %4.0f points \n',size(A,1),
size(A,2));
  fprintf(1, 'No need for grouping, solving as normal \n') ;

  return


elseif(size(A,2) > 10^4)

  fprintf(1,['There are in total ', num2str(size(A,2)), ' pscs in the
network',...
             'process might crash due to memory limit, Good luck!' ]) ;

else

  fprintf(1,'The network has %4.0f arcs and %4.0f points \n',size(A,1),
size(A,2));
  fprintf(1, 'Needs grouping \n') ;
```

```
  fprintf(1, 'Grouping starts ...\n') ;

end

continue_flag = 1 ;

Np_total = size(A,2) ; % total number of points = Npscs
                       % A is transposed

if(strcmp(unwrapping_mode, 'sequential'))

  Ngroup =  ceil(size(A,1)/Np_total) ; % initial

else

  Ngroup =  Nworkers ;

end

while(continue_flag)

  Np_group = floor(Np_total/Ngroup) ;  % number of pscs in the group

  for n = 1 : Ngroup

    if (n < Ngroup)

      [I,J] =  find(A(:,(n-1)*Np_group+1:n*Np_group)~=0) ;
      arc_list = I' ;

    else

      [I,J] =  find(A(:,(n-1)*Np_group+1:Np_total)~=0) ;
      arc_list = I' ;
    end


    % arc_group_all{n} = arc_list ;
    arc_group{n} = unique(arc_list) ; % commom arc_index in the group
                              % avoid multiple times
                              % computation with the normal
                              % matrix later
    Narc_group(n) = length(arc_group{n}) ;

  end

  if(max(Narc_group) <= Np_total)

    continue_flag = 0 ;

  else

    Ngroup = Ngroup + 1 ;

  end

end
N_arc_avg = mean(Narc_group) ;
```